Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI

CONTACT

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TECHNICAL SKILLS

Avionics
MIL-STD-1553 / ARINC 429, Avionics integration

Scientific instrumentation
Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems

Embedded systems
Embedded C, ESP32 / STM32 microcontrollers

Computer vision
Image super-resolution, Object detection

Drones & UAV systems
UAV payload integration

Edge AI
NVIDIA Jetson deployment

AI / machine learning
Deep learning, PyTorch

EDUCATION

Ph.D., Electrical and Electronics Engineering
Northfield Institute of Science and Technology
2021 – ongoing
Thesis: Lightweight deep networks for real-time enhancement of aerial imagery

M.Tech., Avionics
Coastal Institute of Aeronautical Engineering
2010
Grade: CGPA 8.6/10
Thesis: Fault-tolerant data acquisition for airborne test instrumentation

B.E., Electronics and Communication Engineering
Northfield Institute of Science and Technology, Northfield University
2004
Grade: First Class with Distinction

CERTIFICATIONS & TRAINING

Deep Learning for Edge Devices

PROFESSIONAL SUMMARY

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

PROFESSIONAL EXPERIENCE

Scientist E Jul 2016 – Present
Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi
Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.
Tools: Python, PyTorch, MQTT, Node-RED, InfluxDB, Grafana, Docker

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

Additional charge: **Workshop In-charge**, Upper Air Division Workshop Apr 2019 – Present
Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

- Supervise technicians and plan preventive maintenance for workshop test equipment.
- Introduced a job-card and spares-inventory system for instrument repairs and calibration work.

Nodal officer: **Nodal Officer, IoT and AI/ML Initiatives** Jan 2021 – Present

Senior Design Engineer Jun 2004 – Jun 2016
Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru
Integration and qualification testing of airborne electronics and UAV payloads.
Tools: MIL-STD-1553, ARINC 429, LabVIEW, C, MATLAB

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.
- Integrated mission-computer and display units on MIL-STD-1553 avionics buses and built the automated test rig used for bench validation.
- Planned environmental qualification (MIL-STD-810 vibration, temperature and humidity) for line-replaceable units and documented the test results.

Graduate Engineer Aug 2002 – May 2004
Test Systems Group, Meridian Electronics Limited, Bengaluru
Built automated test equipment for avionics line-replaceable units.
Tools: LabVIEW, C

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

Fictional sample profile — not a real person

Example Academy of Applied Computing
Mar 2022

Workshop on Efficient Deep Learning for Embedded Vision
Example Conference on Instrumentation (ECI 2023)
Dec 2023

Introduction to Machine Learning (12-week course)
Example Online University Consortium
Oct 2018

Public Procurement and GFR 2017
Example Institute of Public Administration (sponsored by Kestrel Atmospheric Instrumentation Laboratory)
Nov 2019

National Seminar on Upper-Air Observing Systems
Example Society of Atmospheric Instrumentation
Aug 2021

KEY PROJECTS

Edge super-resolution pipeline for UAV imagery *(Ongoing)* Sep 2021 – Present
Principal investigator, Kestrel Atmospheric Instrumentation Laboratory
Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.
Technologies: NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

Radiosonde ground-station acceptance test protocol Jan 2018 – Mar 2019
Coordinator
Station acceptance testing varied between sites.

- A common acceptance protocol and checklist adopted for new ground-receiving systems.

IoT monitoring of laboratory infrastructure Feb 2020 – Jun 2022
Project lead
Instrument rooms lacked continuous environmental and power-quality monitoring.
Technologies: ESP32, Environmental sensors, MQTT, Node-RED, InfluxDB, Grafana

- Deployed ESP32 sensor nodes publishing over MQTT to a Node-RED flow, InfluxDB time-series storage and Grafana dashboards with threshold alerts.
- Continuous monitoring with automated alerts replaced manual twice-daily readings.

PUBLICATIONS

[1] **A. Sharma** and R. Iyer, "Knowledge-distilled lightweight networks for drone image super-resolution," accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, "Benchmarking embedded GPUs for real-time image enhancement," Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, "Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs," Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, "An IoT telemetry framework for upper-air ground stations," Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, "Edge telemetry dashboards for upper-air ground stations," Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, "A low-cost GNSS receiver chain for radiosonde tracking," Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

MANUSCRIPTS UNDER REVIEW / IN PREPARATION

[1] **A. Sharma**, M. Khan and R. Iyer, "Detection-aware quality assessment of super-resolved drone images," under review at Example Transactions on Geoscience Imaging. *[Under review]*

[2] **A. Sharma**, "Simulation study of balloon-released gliding sondes," in preparation. *[In preparation]*

AWARDS & HONOURS

Best Paper Award 2023
Example Conference on Instrumentation (ECI 2023)

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